

Total No. of Questions : 6]

SEAT No. :

P4868

[Total No. of Pages : 2

T.E./Insem. - 101

T.E. (Civil)

**HYDROLOGY AND WATER RESOURCE ENGINEERING
(2012 Pattern) (Semester - I)**

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Answer Q1 or Q2, Q3 or Q4, Q5 or Q6.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Assume Suitable data if necessary.*

UNIT - I

- Q1)** a) Explain all forms of precipitation. [5]
b) Explain dilution technique for measuring stream discharge. [5]

OR

- Q2)** a) The average annual rainfalls in cm at four existing rainguage stations in a basin are 105, 79, 70 and 66. If the average depth of rainfall over the basin is to be estimated within 10 % error, determine the additional number of gauges needed. [6]
b) State various types of precipitation & explain cyclonic precipitation. [4]

UNIT - II

- Q3)** a) State various advantages of Drip Irrigation. [4]
b) The following data gives details of various crops grown in culturable area of 2000 hectares . Determine the discharge. [6]

Sr.	Crop	Irrigation intensity	Kor period in days	Kor depth in cm
1	Wheat	40%	16	15
2	Jowar	50%	10	12

P.T.O.

OR

- Q4)** a) Explain various factors on which the duty of water for crop depends.[5]
b) State the various methods of assessment of water charges and compare their merits and demerits. [5]

UNIT - III

- Q5)** a) Explain [4]
i) Aquifer
ii) Aquiclude
iii) Specific yield of an aquifer
iv) Porosity of soil
- b) Calculate the diameter of the well that will have a discharge of 300 litre/sec. with a drawdown of 6 m in a unconfined aquifer of thickness of 40 m. The radius of influence is 300 m and the coefficient of permeability is 100 m/day. [6]

OR

- Q6)** a) Discuss the assumptions and limitations of Dupit's theory. [5]
b) Explain with a neat sketch division of subsurface water. [5]

